

Shreya Hirpathak

Engineering Designer · Design Researcher

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SUMMARY

Design researcher and industrial designer (M.S. Engineering Design, Penn State '26) investigating how users' diverse backgrounds and lived experiences shape confidence, decision-making, and interaction with designed systems — published in two ASME IDETC-CIE papers, including a validated 17-item self-efficacy scale built on 1,190+ student responses. Grounded by five product design internships spanning consumer products, packaging, and medical equipment.

EDUCATION

The Pennsylvania State University | M.S., Engineering Design Innovation *Jun 2024 – Aug 2026*

MIT Institute of Design, Pune | B.Des., Industrial Design *Jun 2020 – Jun 2024*

EXPERIENCE

Graduate Student Researcher | The Pennsylvania State University *Nov 2024 – Present*

- Designed and administered a two-round survey to 1,910 first-year engineering students, generating a dataset used to study self-efficacy, decision-making, and makerspace use.
- Ran confirmatory factor analysis (CFA) validating 94% (16 of 17) of items on a self-developed Likert-scale survey; applied item response theory (IRT) to isolate 2 background factors requiring system-level improvement.
- Co-authored a validated 17-item Makerspace Self-Efficacy Scale (MSES), built on responses from 1,190+ engineering students; accepted at ASME IDETC-CIE 2026, won 2nd place Best Paper, and nominated for an ASME special edition journal publication.

Teaching Assistant, EDSGN 100 | The Pennsylvania State University *Aug 2024 – Present*

- Taught design process, prototyping, and CAD (Fusion 360, SolidWorks) to first-year engineering students across 13 sections over 6 semesters, averaging 40 students per class (~520+ students total).
- Delivered lectures and hosted weekly office hours; mentored student design projects and graded coursework for the Engineering Design Innovation department.

Therefore Design, Pune | Product & Industrial Design Intern *Jan 2024 – Jun 2024*

- Completed final-year graduation thesis (MIT Institute of Design capstone) at one of India's leading integrated design consultancies.
- Redesigned a reach-in pharmaceutical stability chamber ("Calidus," for LabIndia) over a 6-month capstone project, benchmarking 10+ global and local competitor products and conducting on-site interviews with factory workers and repair technicians to identify design gaps.
- Reduced assembly process by 67% (from 15 steps to 5) and narrowed unit footprint to 970mm width vs. ~1,300mm on comparable market chambers, while improving ease-of-repair through redesigned dismantling and wire management systems.
- Developed and evaluated 3 concept directions before finalizing the design; built a 1:5 scale model and redesigned a 6-screen HMI/UI system.

Godrej & Boyce (Interio – Seating) | Product Design Intern *May 2023 – Jun 2023*

- Benchmarked 17 competitor gaming chairs across 7 brands, analyzing pricing, materials, and ergonomic features to identify market gaps and design opportunities.
- Designed and circulated a user survey (90 respondents, plus 10 in-depth interviews); found 90.7% played video games regularly, 63% had a home gaming setup, yet 93.6% lacked a gaming chair — insight that shaped Godrej's target positioning.
- Synthesized findings into 4 user personas and 3 design briefs, informing product direction across price tiers (₹8,000–₹25,000).

Guddee (startup) | Product Design Intern *Aug 2022 – Sep 2022*

- Designed products that bring traditional Indian artisan crafts into contemporary daily life, translating handicraft techniques into modern product concepts.

Chef at Home (startup) | Packaging Design Intern *Jun 2022 – Jul 2022*

- Designed launch packaging for "Zangbox," a new brand of cook-it-yourself meal kits curated with five-star hotel chefs.

RESEARCH & PUBLICATIONS

ASME IDETC-CIE 2026 — Houston, TX (accepted) *Aug 2026*

- Hirpathak, S., et al. "Creating a MSES: Developing the Makerspace Self-Efficacy Scale." Paper No. IDETC2026-190931. Developed and validated a 17-item self-efficacy scale via confirmatory factor analysis of 1,190+ engineering student surveys; won 2nd place Best Paper and nominated for an ASME special edition journal publication.

CERS 2026 — Penn State College of Engineering Research Symposium

Apr 2026

- Presented an original academic self-efficacy scale linked to UX research principles.

ASME IDETC-CIE 2025 — Anaheim, CA

Aug 2025

- Presented abstract IDETC2025-169096 on how diverse backgrounds shape design decisions and confidence in makerspaces.

Packaging of the World — Featured Design

Feb 2023

- Sushi take-away packaging for the brand "Ginkgo" featured internationally.

SKILLS & SOFTWARE

Software: Rhino 3D, Fusion 360, Keyshot, Blender, Figma, Framer, Photoshop, Illustrator, Final Cut Pro, Tableau

Design & Research: 3D modeling, prototyping, 3D printing, ideation, user research, survey design, data analytics, confirmatory factor analysis, item response theory

Languages: English, Marathi, Hindi (fluent); German (basic)

ADDITIONAL

- CRY volunteer (2023)
- Bharatanatyam classical dancer — 4-year diploma and arangetram, Tilak Maharashtra Vidyapeeth, Pune (2020).